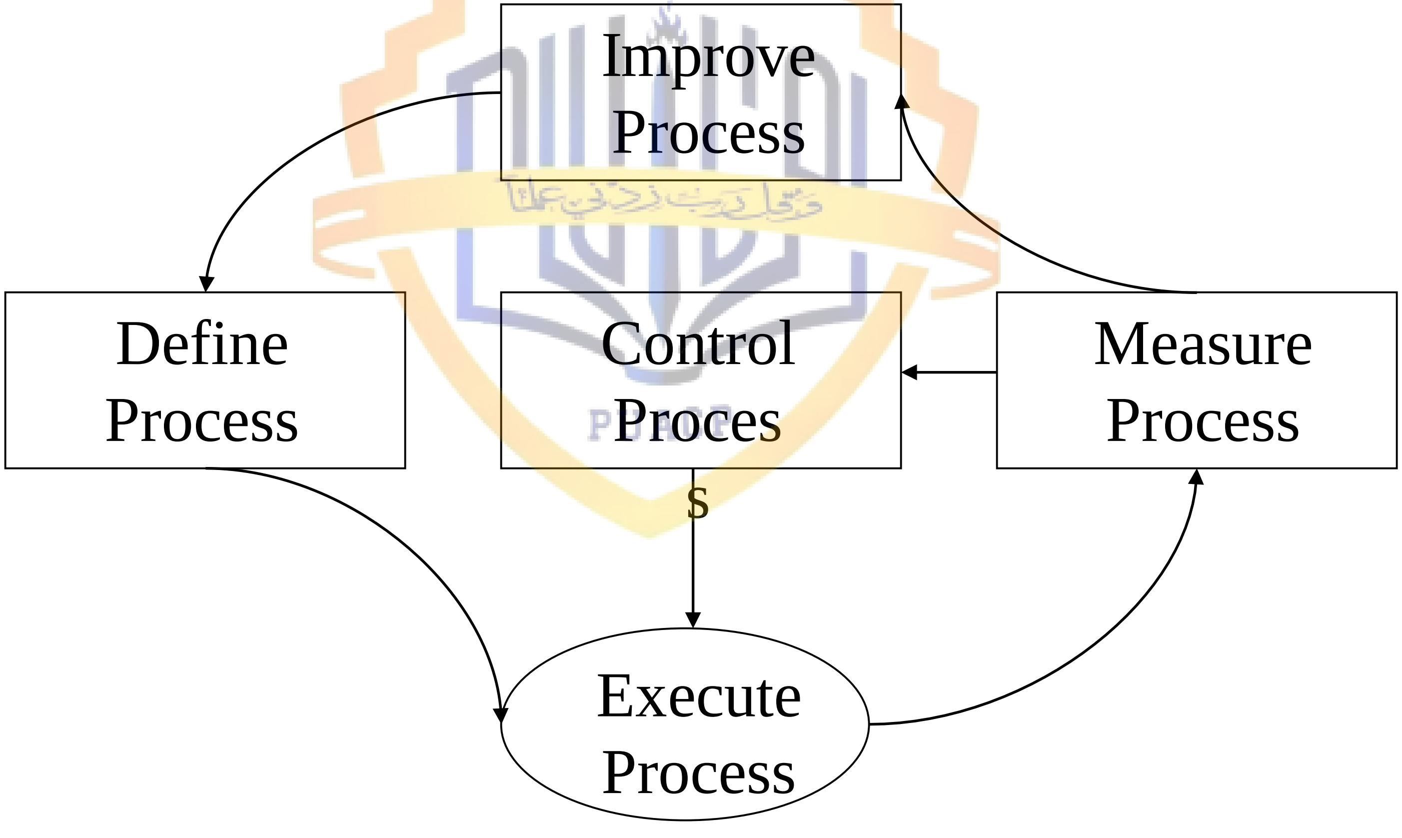
Process Management and Improvement – 1 - CMM

Lecture # 39

Process Management Responsibilities

- Define the process
- Measure the process
- Control the process
  - Ensure variability is stable. Why?
- Improve the process

Process Management Responsibilities



Objectives of Process Definition - 1

- Design processes that can meet or support business and technical objectives
- Identify and define the issues, models, and measures that relate to the performance of the processes
- Provide infrastructures (the set of methods, people, and practices) that are needed to support software activities

Objectives of Process Definition - 2

- Ensure that the software organization has the ability to execute and sustain the processes
  - Skills
  - Training
  - Tools
  - Facilities
  - Funds

Objectives of Process Measurements - 1

- Collect the data that measure the performance of each process
- Analyze the performance of each process

Objectives of Process Measurements - 2

- Retain and use data:
  - To assess process stability and capability
  - To interpret the results of observations and analyses
  - To predict future costs and performance
  - To provide baselines and benchmarks
  - To plot trends
  - To identify opportunities for improvements

Objectives of Process Control - 1

- Controlling a process means keeping within its normal (inherent) performance boundaries – that is, making the process behave consistently

Objectives of Process Control - 2

- Measurement
  - Obtaining information about process performance
- Detection
  - Analyzing the information to identify variations in the process that are due to assignable causes
- Correction
  - Taking steps to remove variation due to assignable causes from the process and to remove the results of process drift from the product

Actions for Process Control - 3

- Determine whether or not the process is under control (is stable with respect to the inherent variability of measured performance)
- Identify performance variations that are caused by process anomalies (assignable causes)

Actions for Process Control - 4

- Estimate the sources of assignable causes so as to stabilize the process

- Once a process is under control, sustaining activities must be undertaken to forestall the effects of entropy. Without sustaining activities, processes can easily fall victim to the forces of ad hoc change or disuse and deteriorate to out-of-control states
- This requires reinforcing the use of defined processes through continuing management oversight, measurement, benchmarking, and process assessments

Objectives of Process Improvement - 1

- Understand the characteristics of existing processes and the factors that affect process capability
- Plan, justify, and implement actions that will modify the processes so as to better meet business needs
- Assess the impacts and benefits gained, and compare these to the costs of changes made to the processes

Objectives of Process Improvement - 2

- Process improvement should be done to help the business – not for its own sake

Identifying Process Issues

- Clarify your business goals or objectives
- Identify the critical processes
- List the objectives for each critical process
- List the potential problem areas associated with the processes
- Group the list of potential problems into common areas or topics

Common Process Issues

- Product quality
- Process duration
- Product delivery
- Process cost

Steps Before Process Improvements - 1

- Explain the problem, discuss why the change is necessary, and spell out the reasons in terms that are meaningful
- Create a comfortable environment where people will feel free to openly voice their concerns and their opinions

Steps Before Process Improvements - 2

- Explain the details of the change, elaborate on the return on investment, how it will effect the staff, and when the change will take place
- Explain how the change will be implemented and measured
- Identify the individuals who are open-minded to accept the change more easily

Steps Before Process Improvements - 3

- Train employees to help them acquire needed skills
- Encourage team work at all times and at all levels
- Address each concern with care so there is no fear left and value each opinion
- Make decisions based on factual data rather than opinions or gut feelings

Steps Before Process Improvements - 4

- Enforce decisions to reinforce the change

Seven Steps of the Process Improvement

- Plan
- Gather data
- Analyze findings
- Describe the ideal process
- Implement the ideal process
- Measure progress
- Standardize the process

Useful Tips on SPI - 1

- Proactively identify and seek support of process-improvement champions and sponsor
- Reinforce management awareness and commitment with a strong business case for each desired process improvement
- Build an infrastructure strong enough to achieve and hold software core competence

Useful Tips on SPI - 2

- Measure the extent of adoption of each desired process improvement until it is effectively, efficiently, and across all appropriate parts of the organization

Process Improvement Programs

- Capability Maturity Model (CMM)
- CMMI
- ISO 9000
- Tick/IT
- Spice
- Total Quality Management (TQM)

What is a Process Model?

- A process model is a structured collection of practices that describe the characteristics of effective processes
- Practices included are those proven by experience to be effective

How is a Process Model Used?

- A process model is used
  - To help set process improvement objectives and priorities
  - To help ensure stable, capable, and mature processes
  - As a guide for improvement of project and organizational processes
  - With an appraisal method to diagnose the state of an organization’s current practices

Why is a Process Model Important?

- A process model provides
  - A place to start improving
  - The benefit of a community's prior experiences
  - A common language and a shared vision
  - A framework for prioritizing actions
  - A way to define what improvement means for an organization

- Process improvement should be done to help the business – not for its own sake
- In God we trust, all others bring data
  - W. Edwards Deming

Capability Maturity Model

Software State-of-the-Art in 1984 - 1

- More than half of the large software systems were late in excess of 12 months
- The average costs of large software systems was more than twice the initial budget
- The cancellation rate of large software systems exceeded 35%
- The quality and reliability levels of delivered software of all sizes was poor

Software State-of-the-Art in 1984 - 2

- Software personnel were increasing by more than 10% per year
- Software was the largest known business expense which could not be managed

Software Engineering Institute

- A research facility, located in University of Carnegie Mellon, Pennsylvania
- Primarily funded by US DoD to explore software issues, and especially topics associated with defense contracts
- US DoD is the largest producer and consumer of software in the world

Capability Maturity Model

- SEI developed a Capability Maturity Model (CMM) for software systems and an assessment mechanism
- CMM has five maturity models
  - Initial
  - Repeatable
  - Defined
  - Managed
  - Optimizing

The Five Levels of Software Process Maturity

1. Initial
Unpredictable and poorly controlled

Disciplined Process

2. Repeatable
Can repeat previously mastered tasks

Project Management

Standard, Consistent Process

3. Defined
Process characterized, fairly well understood

Integrated Engineering Process

Predictable Process

4. Managed
Process measured and controlled

Product and Process Quality

Continuously Improving Process

5.Optimizing
Focus on process improvement

Managing Change

CMM Level 1: Initial - 1

- Organizations are characterized by random or chaotic development methods with little formality and uninformed project management
- Small projects may be successful, but larger projects are often failures
- Overall results are marginal to poor

CMM Level 1: Initial - 2

- In terms of People CMM, level 1 organizations are deficient in training at both the technical staff and managerial levels

CMM Level 1: Initial - 3

- SEI does not recommend any key process areas

CMM Level 2: Repeatable - 1

- Organizations have introduced at least some rigor into project management and technical development tasks
- Approaches such as formal cost estimating are noted for project management, and formal requirements gathering are often noted during development
- Compared to initial level, a higher frequency of success and a lower incidence of overruns and cancelled projects can be observed

CMM Level 2: Repeatable - 2

- In terms of People CMM, level 2 organizations have begun to provide adequate training for managers and technical staff
- Become aware of professional growth and the need for selecting and keeping capable personnel

CMM Level 2: Repeatable - 3

- Key process areas
  - Requirements management
  - Software project planning
  - Software project tracking and oversight
  - Software subcontract management
  - Software quality assurance
  - Software configuration management

CMM Level 2: Repeatable - 4

- Key process areas for People CMM
  - Compensation
  - Training
  - Staffing
  - Communication
  - Work environment

CMM Level 3: Defined - 1

- Organizations have mastered a development process that can often lead to successful large systems
- Over and above the project management and technical approached found in Level 2 organizations, the Level 3 groups have a well-defined development process that can handle all sizes and kinds of projects

CMM Level 3: Defined - 2

- In terms of People CMM, the organizations have developed skills inventories
- Capable of selecting appropriate specialists who may be needed for critical topics such as testing, quality assurance, web mastery, and the like

CMM Level 3: Defined - 3

- Key process areas
  - Organization process focus
  - Organization process definition
  - Training
  - Software product engineering
  - Peer reviews
  - Integrated software management
  - Inter-group coordination

CMM Level 3: Defined - 4

- Key process areas for People CMM
  - Career development
  - Competency-based practices
  - Work force planning
  - Analysis of the knowledge and the skills needed by the organization

CMM Level 4: Managed - 1

- Organizations have established a firm quantitative basis for project management and utilize both effective measurements and also effective cost and quality estimates

CMM Level 4: Managed - 2

- In terms of People CMM, organizations are able to not only monitor their need for specialized personnel, but are actually able to explore the productivity and quality results associated from the presence of specialists in a quantitative way
- Able to do long-range predictions of needs
- Mentoring

CMM Level 4: Managed - 3

- Key process areas
  - Software quality management
  - Quantitative software management

CMM Level 4: Managed - 4

- Key process areas for People CMM
  - Mentoring
  - Team building
  - Organizational competency
  - Ability to predict and measure the effect of specialists and teams in quantitative manner

CMM Level 5: Optimizing - 1

- Organizations are assumed to have mastered the current state-of-the-art of software project management and development

CMM Level 5: Optimizing - 2

- In terms of People CMM, the requirements are an extension of the Level 4 capabilities and hence different more in degree than in kind
- Stresses both coaching and rewards for innovation

CMM Level 5: Optimizing - 3

- Key process areas
  - Defect prevention
  - Technology change management
  - Process change management

CMM Level 5: Optimizing - 4

- Key process areas for People CMM
  - Encouragement of innovation
  - Coaching
  - Personal competency development

Levels/ Process Categories	Management	Organizational	Engineering
5 Optimizing		Technology Change Management Process Change Management	Defect Prevention
4 Managed	Quantitative Software Management		Software Quality Management
3 Defined	Integrated Software Management Intergroup Coordination	Organization Process Focus Organization Process Definition Training Program	Software Product Engineering Peer Reviews
2 Repeatable	Requirements Management Software Project Planning Software Project Tracking and Oversight Software Subcontract Management Software Quality Assurance Software Configuration Management		
1 Initial		Ad Hoc Processes	

Level 1 Quality

- Software defect potentials run from 3 to more than 15 defects per function points, but average is 5 defects per function point
- Defect removal efficiency runs from less than 70% to more than 95%, but average is 85%
- Average number of delivered defects is 0.75 defects per function point
- Several hundred projects surveyed

Level 2 Quality

- Software defect potentials run from 3 to more than 12 defects per function points, but average is 4.8 defects per function point
- Defect removal efficiency runs from less than 70% to more than 96%, but average is 87%
- Average number of delivered defects is 0.6 defects per function point
- Fifty (50) projects surveyed

Level 3 Quality

- Software defect potentials run from 2.5 to more than 9 defects per function points, but average is 4.3 defects per function point
- Defect removal efficiency runs from less than 75% to more than 97%, but average is 89%
- Average number of delivered defects is 0.47 defects per function point
- Thirty (30) projects surveyed

Level 4 Quality

- Software defect potentials run from 2.3 to more than 6 defects per function points, but average is 3.8 defects per function point
- Defect removal efficiency runs from less than 80% to more than 99%, but average is 94%
- Average number of delivered defects is 0.2 defects per function point
- Nine (9) projects surveyed

Level 5 Quality

- Software defect potentials run from 2 to 5 defects per function points, but average is 3.5 defects per function point
- Defect removal efficiency runs from less than 90% to more than 99%, but average is 97%
- Average number of delivered defects is 0.1 defects per function point
- Four (4) projects surveyed

Summary

References

- Software Quality: Analysis and Guidelines for Success by Capers Jones
- Measuring the Software Process by William Florac and Anita Carleton, (Chapters 1 & 2)
- The Capability Maturity Model: Guidelines for Improving the Software Process, by SEI, Pearson Education, 2002